

Supplementary Information

The emergence of sharing networks through indirect signaling

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1 Network signaling under uncertainty: the full model

Consider a population of N agents linked by an exogenous communication network whose degree distribution has finite mean $\bar{k}$. A single *focal giver* i in the population dispenses aid to a sequence of *receivers* $j = 1, 2, \dots$, for a total number of receivers k . Each receiver emits a short-lived signal about i 's generosity that propagates one step along the communication network to the receiver's neighbors, whom we call *targets*. A target τ that hears the signal grants the giver a reputational payoff $\varphi(s_{i,\tau})$. Assume that the social reward function can be written as an exponential surprise decay:

$$\varphi(s_{i,\tau}) = e^{-\beta s_{i,\tau}}, \quad s_{i,\tau} : \text{number of previous signals } \tau \text{ has heard about } i. \quad (1)$$

Hence $\varphi(0) = 1$ and $\beta > 0$ captures *diminishing surprise*, where the marginal reward a target confers to i decreases with every additional signal they get. Thus, reputational rewards are a concave function of signals received. $\beta = 0$ is the case of constant surprise where marginal social rewards remain constant, and $\beta < 0$ is the case of inflating surprise where marginal social rewards *increase* with every additional signal (convex reputational rewards).

The focal giver only knows that there is a probability u that any particular receiver has a connection to any other community individual—a target. Their problem becomes then one of choosing *how many* receivers k to share with, given the cost of sharing. Let

$$u = \frac{\bar{k}}{N-1} \in (0, 1) \quad (2)$$

denote the network density—the probability that two randomly chosen agents in the communication network are directly connected, excluding the focal giver. We assume independent mixing so that each target's links to the first $j-1$ receivers follow a Binomial($j-1, u$) distribution (resulting in an Erdős-Rényi random network). Let $r = (1-u) + u e^{-\beta} = 1 - u(1 - e^{-\beta})$ and note that $r < 1 \iff \beta > 0$. Then for the j -th receiver the expected reputational benefit is

$$\mathbb{E}[W_j] = (N-1) u \sum_{m=0}^{j-1} \binom{j-1}{m} u^m (1-u)^{j-1-m} e^{-\beta m} \quad (3)$$

$$= (N-1) u r^{j-1} \quad (4)$$

$$= \bar{k} r^{j-1} \quad (5)$$

The Binomial factor in (3) accounts for the m earlier receivers who might already have been connected to the same target. Summing the geometric series gives

$$\sum_{j=1}^k \mathbb{E}[W_j] = w(k; u, \beta) = \bar{k} \frac{1 - r^k}{1 - r}, \quad k = 1, 2, \dots \quad (6)$$

The giver starts with surplus resources h and pays a cost $C(k; c) = k c$ for maintaining k connections, so the expected utility after sharing with k receivers is

$$V(k; u, \beta) = w(k; u, \beta) - k c, \quad 0 \leq k \leq k_{\max} = \left\lfloor \frac{h}{c} \right\rfloor. \quad (7)$$

Define $\Delta w(k) = w(k) - w(k-1)$. We have two possible cases:

1. *Diminishing surprise* ($\beta > 0$, $0 < r < 1$). In this case the social reward curve is strictly concave, and there can be a $k^* \in [0, k_{\max}]$. The realized number of receivers then is

$$\hat{k} = \min\{k^*, k_{\max}\}. \quad (8)$$

Wealthier givers (h large) thus can afford more links, but even they stop at the point where marginal reputation equals marginal cost.

2. *Constant or increasing surprise* ($\beta \leq 0$, $r \geq 1$). Each new receiver always yields the same (or larger, in the case that $\beta < 0$) marginal payoff $\Delta w = \bar{k} r^{k-1}$ since in such cases $r \geq 1$; if $\Delta w > c$ the giver pushes to the budget limit $\hat{k} = k_{\max}$, otherwise $\hat{k} = 0$.

We can calculate k^* by solving $\Delta w(k^*) = c$ for k^* . This gives

$$k^* = \begin{cases} \left\lfloor 1 + \frac{\log[c/\bar{k}]}{\log r} \right\rfloor, & c < \bar{k}, \\ 0, & c \geq \bar{k} \end{cases} \quad (9)$$

where $\log$ refers to the natural base e logarithm.

For further insight, we will go to the limits of β . First, we evaluate $\lim_{\beta \rightarrow \infty} w(k)$. This means we treat reputational payoffs as if they came mostly or completely from the initial surprise a target receives from hearing about the giver for the first time:

$$\lim_{\beta \rightarrow \infty} w(k) = (N-1) [1 - (1-u)^k], \quad (10)$$

with an optimal number of connections given by

$$k^*(u, \infty) = \begin{cases} \left\lfloor 1 + \frac{\log[c/\bar{k}]}{\log(1-u)} \right\rfloor, & c < \bar{k}, \\ 0, & c \geq \bar{k}, \end{cases} \quad (11)$$

leading to a social reward given by the total community size, $N-1$, weighted by the probability that any particular target has gotten a signal from any of the k receivers. If this probability is low ($u \approx 0$) and $\bar{k} > c$ then k^* can be large, as seeking out many sharing partners becomes essential for saturating the sparse network with the signal. But as u decreases so does $\bar{k}$, to the point that it crosses the threshold $\bar{k}_{\min} = c$. At this point it is not worthwhile to share anymore and $k^* = 0$. If, on the other hand, the communication network is dense ($u \approx 1$) then the incentive to form many connections also decreases, as only a few connections are able to saturate the network with i 's signal without the need to pay for more. If $\bar{k} > c$ and $u \rightarrow 1$, it becomes worthwhile to make exactly $k^* = 1$ connection, and otherwise none.

We can also take the limit as β approaches 0 from the right, in order to examine the case of constant surprise. In this case, as mentioned above, the benefits obtained from signaling do not decay with repeated signals. By L'Hôpital's rule when $\beta \rightarrow 0$ from the right:

$$\lim_{\beta \rightarrow 0^+} w(k) = k \bar{k}, \quad (12)$$

and the optimal number of connections is approximately

$$k^*(u, 0^+) \approx \begin{cases} \left\lfloor 1 - \frac{\log[c/\bar{k}]}{u\beta} \right\rfloor, & c < \bar{k}, \\ 0, & c \geq \bar{k}, \end{cases} \quad (13)$$

If getting one connection is profitable, which is represented by the condition $\bar{k} > c$, then agents find it advantageous to be givers, and they are incentivized to give out to as many receivers as they possibly can, which in Equation 13 can be seen as the fractional term on the right hand side quickly grows as β goes to zero. Otherwise, no outward connections are made.

Under diminishing surprise, sharing is worthwhile only when potential receivers are sufficiently well connected to propagate the signal into new parts of the network, but not so connected that sharing widely is disincentivized, especially if social rewards saturate quickly. Sharing with someone with little-to-no connections yields no benefits from reputation-building, and minimal sharing will provide most of the benefit in a dense network, especially in one that does not value repeated signals.

Endowment heterogeneity matters because it determines the personal budget cap $k_{\max} = \lfloor h/c \rfloor$; the optimal number of connections is then

$$\hat{k} = \min\{k^*, \lfloor h/c \rfloor\}, \quad (14)$$

such that under this simple model, richer individuals have a higher connection budget. Under constant surprise, the optimal number of connections is either h/c or 0. Trivially, in case of increasing surprise this result would remain the same (since $\beta < 0$ just makes marginal benefits grow faster; optimal k still saturates at the budget.)

The case of diminishing surprise allows us to see an important part of sharing and indirect signaling in plain mathematics: the utility one gets from sharing depends on the probability that the people one shares with have the connections to effectively spread reputation-building signals after being shared with. Even if giving to the needy is socially rewarded, if only indirect signals are allowed as reputation-building mechanisms, then it will not be worthwhile to share with individuals who are not sufficiently connected—regardless of their need status—so long as there are better-connected individuals who are in need and willing to indirectly reciprocate. In a community that rewards generosity, those who are not well-connected in the community's communication network can thus “fall through the cracks” of informal social aid. By this same logic, the emergence of inequality in sharing patterns is incentivized in communities with sparse communication networks, as in such networks better-connected individuals are more scarce and also more effective vectors for spreading a good reputation relative to the average individual. In communities with dense communication networks, on the other hand, the benefit of sharing with any particular individual is likelier to be more homogenous, as the target sets of most potential receivers will tend to overlap considerably.

2 The general network signaling model

The above game can be characterized as a special case of a more general network signaling model. Such a general framework is itself a case of a well-studied class of games in the submodular optimization literature. Even if there is no real novelty in specifying this general model, we include it here for curious readers, and refer to the appropriate literature at the end of this section.

Formally, we define the game as follows:

- $G = (V, E)$: fixed communication network with V the set of all nodes in the network and $|V| = N$ the total number of nodes. E is the set of edges so that $|E|$ is the total number of edges.
- $i \in V$: focal giver.
- k : number of distinct receivers chosen by i , indexed $j = 1, \dots, k$.
- $\tau \in V \setminus \{i, j\}$: potential targets reached by sharing with j (excluding focal giver and receiver).
- $A_{j,\tau} \in \{0, 1\}$: adjacency matrix indicating whether receiver j is directly linked to target τ in G .
- $s_{i,\tau}$: number of prior signals a target τ has heard about the focal giver.
- $\varphi(s)$: *general social-reward kernel*, a function $\varphi : \mathbb{N}_0 \rightarrow \mathbb{R}_{\geq 0}$ such that $\varphi(s)$ is the social reward received by a focal signaler for targeting an individual who has already received s signals from other senders in the current round.

Before the j -th gift, target τ has heard

$$s_{i,\tau}^{(j)} := \sum_{m=1}^{j-1} A_{m\tau} \quad (15)$$

signals about the giver. The marginal reputational gain from gift j is

$$w_j(G, \Theta_w) = \sum_{\tau \in V} A_{j\tau} \varphi(s_{i,\tau}^{(j)}; \Theta_w) \quad (16)$$

with Θ_w a set of parameters governing the benefits of network signaling. Aggregating over gifts gives

$$W(k; G, \Theta_w) = \sum_{j=1}^k \sum_{\tau \in V} A_{j\tau} \varphi(s_{i,\tau}^{(j)}, \Theta_w). \quad (17)$$

It is convenient to write this solely in terms of the *total* number of signals each target receives:

$$s_{i,\tau} := \sum_{j=1}^k A_{j\tau}, \quad (18)$$

then re-index the inner sum to obtain

$$\boxed{W(k; G, \Theta_w) = \sum_{\tau \in V} \sum_{m=0}^{s_{i,\tau}-1} \varphi(m; \Theta_w)} \quad (19)$$

Equation 19 separates network structure (via the $s_{i,\tau}$ counts) from cultural and psychological preference assumptions (through the kernel φ).

Let $C(k; \Theta_C)$ be the cost of k gifts and Θ_C a set of cost parameters. Given any network G and kernel φ , the discrete optimization problem is

$$k^* := \arg \max_{0 \leq k} \{W(k; G, \Theta_w) - C(k; \Theta_C)\}. \quad (20)$$

In practice one seeks analytical insight by imposing structure on G (e.g., random graphs or degree-regular graphs), φ (e.g., exponential, leading to closed-form expressions shown in the next section), and C (e.g., linear). With complete information on G , choosing a receiver set $S \subseteq V$ yields one-hop coverage $f(S) = |\bigcup_{j \in S} N_{\text{com}}(j)|$ (where $N_{\text{com}}(j)$ is the set of communication partners of j), which is monotone submodular. If each target τ gives a payoff with a concave, non-decreasing function g of its exposure count $s_{i,\tau}(S)$, then $F_i(S) = \sum_{\tau} g[s_{i,\tau}(S)]$ is monotone submodular. Maximizing $F_i(S)$ under a cardinality constraint $|S| \leq k_{\max}$ admits the classic greedy $1 - \frac{1}{e}$ guarantee [5], while under a knapsack budget $\sum_{j \in S} c_j \leq B$ naïve density-greedy does not in general achieve $1 - \frac{1}{e}$; standard refinements (e.g., partial enumeration plus greedy, or continuous-greedy with rounding) do recover $(1 - \frac{1}{e} - \epsilon)$ guarantees (see [4] for a contemporary review). Diminishing returns come from submodularity; in the sequential k -choice view, a unique single-crossing arises when marginal benefits decrease in k (submodularity) and marginal costs are nondecreasing (linear or convex).

3 Agent-based model, dataset and ABC-PMC algorithm

3.1 Strategic network formation algorithm

Our simulation model follows a population of agents, each assigned an endowment h_i and a private store of reputation signals and payoffs. Agents sit on two networks:

1. **Communication graph** (fixed, undirected). At initialization we sample an Erdős–Rényi network of N nodes with link probability u . If $u < 1$, we then run an iterative rewiring procedure for a set number of steps (1000 by default), where a single edge addition and removal are done per step, and they are biased by a transformed endowment-based weight so that nodes closer to a reference endowment size $\zeta \in [0, \max(h)]$ tend to be preferential targets for edge additions and are avoided for edge removals, thus ending up with higher degree. The relationship between inverse distance to the reference endowment and communication network degree can be made stronger or weaker by tuning of the strength parameter $S \geq 0$. If $S > 0$, an individual with endowment that is close to ζ has a higher probability of gaining more edges in the communication network than an individual with an endowment size further away from ζ . This probability increases with S . When $S = 0$, the network is randomly rewired, recovering an Erdős–Rényi graph of density u . If $u = 1$, we forego this whole procedure and simply use the complete graph with N vertices. This gives each agent its initial communication neighbors and communication degree while keeping the communication network density constant and given by u .
2. **Sharing graph** (dynamic, directed). Starts empty and grows as agents form outgoing ties in search of social rewards.

This means that the communication layer is fixed at the moment in which the sharing layer is generated. To generate the sharing layer, at each time step we shuffle the agents and give each one the opportunity to form a single new outgoing tie. They pick a random stranger, then calculate the

extra social reward they would harvest—by summing up reputation signals flowing through that candidate’s communication neighbors—and subtract a cost that rises with their number of links (and depends on their own harvest). If the net gain beats what their current payoff, they add the edge, update both agents’ payoffs and reputation scores, and end their turn. If adding the tie does not improve their payoff, they do nothing.

At the final time step we log every agent’s in- and out-degree, local clustering, PageRank, coreness, and compute population-level summaries: mean degree, median in- and out-degree, average clustering coefficient, average coreness, count of zero in-degree nodes, correlation between PageRank and endowment, and Gini coefficient on in-degrees. These statistics provide descriptions of aspects of the empirical sharing network structure that we want the model to be able to emulate (see Inference Algorithm, below).

The payoffs which agents use to make sharing decisions are given as follows. When agent i shares with partner j , j emits a signal to each of its communication-neighbors τ . Each such τ keeps a cumulative count of all of i ’s sharing links that it has ever heard of:

$$s_{i,\tau} = \sum_{n \in N_{\text{com}}(\tau)} \mathbf{1}\{(i \rightarrow n) \in E_{\text{share}}\} = |K_i \cap N_{\text{com}}(\tau)|, \quad (21)$$

where K_i is the set of out-neighbors of i in the sharing graph.

To turn that history into a reputation payment, we use the same functional form as the analytical model above. τ treats each of its $s_{i,\tau}$ signals as an exponentially-discounted utility token $\varphi(s) = e^{-\beta s}$, which we sum up to obtain a suitable cumulative social reward function:

$$r_{\tau \rightarrow i} = \sum_{m=0}^{s_{i,\tau}-1} e^{-\beta m} = \frac{1 - e^{-\beta s_{i,\tau}}}{1 - e^{-\beta}}. \quad (22)$$

for $\beta > 0$. Incorporating the $\beta = 0$ case gives the piecewise form

$$r_{\tau \rightarrow i} = \begin{cases} \frac{1 - e^{-\beta s_{i,\tau}}}{1 - e^{-\beta}}, & \beta > 0, \\ s_{i,\tau}, & \beta = 0, \end{cases} \quad (23)$$

where β , as before, controls the curvature of the social reward function, with smaller β leading to slower saturation.

Agent i may have multiple sharing-partners $j \in K_i$, and any target τ could lie in several of their neighborhoods. To avoid double-counting, we form the union of all those neighborhoods and sum each τ exactly once:

$$R_i(K_i) = \sum_{\tau \in \bigcup_{j \in K_i} N_{\text{com}}(j)} r_{\tau \rightarrow i}. \quad (24)$$

In order to capture effects of endowment dependence and repeated sharing on the scaling of sharing costs, we relax the linear cost assumption of the analytical model in the previous section and instead let costs scale with k_i and resource endowment (harvest) $h_i = H_i + 1$, where H is the (zero inflated) empirical distribution of harvest sizes in the empirical data (see Dataset). Maintaining $k_i = |K_i|$ outgoing links thus incurs a cost

$$C_i(k_i; c, \ell, \gamma) = c \frac{k_i^\ell}{h_i^\gamma}, \quad (25)$$

so that links become marginally pricier whenever $\ell > 1$ (convex costs), increase linearly with $\ell = 1$ and become marginally cheaper when $\ell < 1$. Agents with larger harvest h_i enjoy a relative cost advantage whenever $\gamma > 0$, and costs are scaled by c .

Finally, the net payoff is simply the positive part of reward minus cost:

$$\pi_i = \max\{0, R_i - C_i\}, \quad (26)$$

ensuring that an agent only keeps or forms ties when the extra, exponentially-filtered endorsements more than cover their upkeep.

All this implies we add an edge $i \rightarrow v$ whenever

$$\Delta R_i(i \rightarrow v) \geq \Delta C_i(i \rightarrow v), \quad (27)$$

with

$$\Delta R_i(i \rightarrow v) = R_i(K_i \cup \{v\}) - R_i(K_i), \quad \Delta C_i(i \rightarrow v) = c \frac{(k_i + 1)^\ell - (k_i)^\ell}{h_i^\gamma}. \quad (28)$$

3.2 Inference algorithm

We wish to infer the joint posterior over the seven key parameters

$$\theta = (\ell, c, \gamma, \beta, u, \zeta, S) \quad (29)$$

given an observed vector of network summaries s_{obs} . To do so we employ a Population Monte Carlo approximate Bayesian computation (ABC-PMC) algorithm with N particles and R replicates per candidate [11; 1; 6].

In this parameterization, ℓ controls the curvature of link-maintenance costs: it appears in $C_i(k_i; c, \ell, \gamma) = c k_i^\ell / h_i^\gamma$, so $\ell > 1$ implies convex costs (each additional tie is increasingly expensive), $\ell = 1$ gives linear costs, and $\ell < 1$ makes marginal costs decline with degree. The parameter c is a global cost scale that sets the overall difficulty of maintaining sharing ties. The exponent γ captures how strongly resource endowment h_i shields an agent from costs: when $\gamma > 0$, richer households face lower effective per-tie costs, amplifying endowment-based advantages in forming and keeping links. The parameter β governs the curvature of reputational rewards via the exponential surprise kernel $\varphi(s) = e^{-\beta s}$: larger β means faster saturation of reputational payoffs with repeated signals, whereas $\beta \approx 0$ corresponds to nearly constant marginal reputational gains. The parameter u is the baseline density of the communication graph (the Erdős-Rényi link probability before rewiring), controlling how easily signals propagate through the community. The reference point ζ sets the endowment level that is favoured when rewiring the communication network: nodes whose endowments lie closer to ζ are more likely to gain communication edges and less likely to lose them. Finally, S controls the strength of this degree-endowment targeting: when $S = 0$ the communication network is effectively random with density u , and as S increases, degree becomes more tightly concentrated around the endowment level ζ .

We assume independent priors

$$\ell \sim \text{Exp}(\lambda_1), \quad c \sim \text{Exp}(\lambda_2), \quad \beta \sim \text{Exp}(\lambda_3), \quad (30)$$

and

$$\gamma, u, \zeta, S \sim \text{Uniform}(0, 1). \quad (31)$$

and set $\lambda_1 = \lambda_2 = 1$, and $\lambda_3 = 0.1$. In the first round we draw $N = 1000$ parameter vectors $\{\theta_i\}$ by Latin-Hypercube sampling from these marginals.

For each candidate θ_i , we run R independent replicas of the ABM for 1000 ticks. At the end of each run we collapse the resulting sharing network into the 8-dimensional summary

$$s(\theta_i) = \begin{bmatrix} \text{med}(d_{\text{in}}), & \text{med}(d_{\text{out}}), & \text{Var}(d_{\text{in}}), & \text{MAD}(d_{\text{out}}), \\ Z_{\text{in}}, & \xi, & \rho_{\text{pr},h}, & G_{\text{in}} \end{bmatrix}, \quad (32)$$

where every measure is chosen for its contribution to the overall sharing network trends we wish to capture:

med(d_{in}) and med(d_{out}) (median in-degree and out-degree). They anchor the overall density of giving and receiving. If they cannot be matched, the core tradeoff is misspecified. med(d_{out}) pins down the *typical* giver’s capacity and prevents fits driven by a few extreme donors, and med(d_{in}) likewise pins down the modal degree of receiving.

Var(d_{in}) and MAD(d_{out}) (variance of in- and median absolute deviation of out-degree). These measures capture heterogeneity in giving strategies. Matching them ensures the model does not produce an unrealistically narrow (or excessively heavy-tailed) donor and/or receiver distribution. Because the Kangiqsujuaq sharing network has a heavily skewed out-degree distribution, we use the median absolute deviation as a measure of dispersion.

Z_{in} (count of zero-in-degree nodes). Measures *coverage gaps*—households that do not receive any transfers in the sharing network. In this mechanism, zeros persist when nodes that are peripheral (in the communication network) offer little marginal audience relative to cost. Matching Z_{in} is thus important for assessing undersupply at the margins.

ξ (clustering coefficient of the sharing network). Diagnoses the *audience-overlap* mechanism. Targeting receivers whose neighborhoods overlap closes triangles in the sharing graph even without a “copy friends of friends” rule. Low simulated ξ with matched $\bar{d}$ (density of the sharing network) suggests missing overlap (e.g., wrong network density, u , or unrealistic assortativity in the communication network).

$\rho_{\text{pr},h}$ (Spearman’s correlation of sharing-PageRank with resource endowment). Key *directionality* diagnostic. PageRank is a measure of centrality that emphasizes endorsements from central givers. Because our empirical data is unweighted, we consider to be a better proxy of significant resource transfers than raw in-degree (see Dataset, below). Our mechanism predicts a negative association when low-endowment households are preferential receivers due to their centrality in the communication layer. A strong positive $\rho_{\text{pr},h}$ would undermine alignment between reputational incentives and need, instead reflecting transfers among a resource-rich and communication-central network core.

G_{in} (Gini of in-degree distribution). Summarizes inequality in received help. Network signaling predicts concentration of sharing toward high-communication centrality nodes as reputational returns and cost convexity more strongly favor selective portfolios. Hitting this target ensures the model captures not only mean coverage but its distribution; it is also responsive to counterfactuals that reduce costs to peripheral nodes (which could lower in-degree inequality, G_{in} , while maintaining the correlation between sharing and resource endowment $\rho_{\text{pr},h} < 0$).

We compare these summary statistics in our simulated networks to the observed s_{obs} computed from the empirical dataset. By fitting to this blend of connectivity (degrees), clustering, centralization (PageRank–endowment correlation), and inequality (degree variance & Gini), we ensure our ABC inference is sensitive to both microscopic link-formation rules and macroscopic network structure.

After a small pilot batch ($5N$ simulations) we estimate the covariance matrix Σ of the summaries and calculate a distance metric for candidate parameter combination θ_i :

$$d(\theta_i) = \sqrt{(s(\theta_i) - s_{\text{obs}})^T \Sigma^{-1} (s(\theta_i) - s_{\text{obs}})}. \quad (33)$$

We choose an initial cutoff $\varepsilon_1 = 3$. This value was chosen based on pilot runs to keep a balance between tolerance looseness and speed during the first round, and otherwise does not affect final results. A candidate θ_i is accepted if its distance satisfies $d(\theta_i) \leq \varepsilon_1$. A round runs the algorithm until $N = 2000$ proposed parameter combinations fall within the threshold and get accepted.

For each subsequent round $t = 2, 3, \dots$, we set

$$\varepsilon_t = Q_\alpha(\{d(\theta_i)\}), \quad (34)$$

the α -quantile of the distances from round $t - 1$ (in the main analyses we use $\alpha = 0.2$). We then form a new batch of N proposals by resampling particles from the previous round $\{\theta_j^{(t-1)}\}$ with probabilities $w_j^{(t-1)}$ and perturbing each parameter independently with a truncated Gaussian kernel. Let $\widehat{\text{Var}}_{t-1}(\theta_k)$ denote the marginal variance of component k in the previous population. For a resampled parent $\theta_j^{(t-1)}$ we draw

$$\theta'_k \sim \begin{cases} \mathcal{N}(\theta_{j,k}^{(t-1)}, \sigma_{k,t}^2) \text{ truncated to } (0, \infty), & k \in \{\ell, c, \beta\}, \\ \mathcal{N}(\theta_{j,k}^{(t-1)}, \sigma_{k,t}^2) \text{ truncated to } (0, 1), & k \in \{\gamma, u, \zeta, S\}, \end{cases}$$

with

$$\sigma_{k,t}^2 = a^2 \widehat{\text{Var}}_{t-1}(\theta_k), \quad a \approx 0.9.$$

Thus each parameter is perturbed with independent Gaussian noise, scaled by the empirical spread of that parameter in the previous round and truncated to its prior support. The constant a controls the overall step size: values that are too small lead to very slow exploration, while values that are too large cause excessive rejection and rapid weight degeneracy. We chose $a \approx 0.9$ based on pilot runs to obtain a balance between acceptance rate and ESS; in practice, the ABC-PMC algorithm only requires that the perturbation variance be proportional to the empirical variance. For each proposed $\theta_i^{(t)}$ we run the agent-based model, compute its Mahalanobis distance $d(\theta_i^{(t)})$, and accept it into round t if $d(\theta_i^{(t)}) \leq \varepsilon_t$, continuing until N particles have been accepted.

Each newly accepted θ_i in round t receives an importance weight

$$w_i^{(t)} \propto \frac{\pi(\theta_i)}{\sum_j w_j^{(t-1)} K_t(\theta_i | \theta_j)} \quad (35)$$

where $\pi(\theta_i)$ is the prior density. We then normalize so $\sum_i w_i^{(t)} = 1$.

We repeat until effective sample size falls below a threshold (we choose the fractional threshold $\text{ESS}/N = 0.3$, excluding rounds below this threshold from the pooled posterior) or once particle

acceptance rates have fallen below 0.5%. As a conservative measure, we ran the ABC-PMC algorithm with five different seeds, and we pool the last round of every seeded run, re-weighting the pooled particles using an exponential kernel $w_i^{\text{pooled}} = \eta \exp(-\frac{1}{2}d(\theta_i))$ where η is a normalizing constant. The final weighted ensemble $\{(\theta_i, w_i^{\text{pooled}})\}$ provides an approximate posterior $p(\theta | s_{\text{obs}})$ over the model parameters with a total of 10000 pooled particles.

3.3 Dataset

The data that we use to inform our analysis were collected by Ready in Kangiqsujuaq, an Inuit community in the Canadian Arctic, between September 2013 and July 2014 as part of a community-based research project focused on food security and economic change (see [7; 10; 8; 9; 2; 3]). A key feature of Inuit communities is the persistence of hunting and gathering as an important part of local livelihoods, despite the transition to permanent settlement and a wage-based economy. Hunted, fished, and gathered “country” foods (locally-harvested foods like caribou, seal, and Arctic char) are still distributed in communities through sharing networks. Local sharing norms are often framed in terms of need-based transfers (i.e., “we try to share to those in need.”)

The network data were collected in a household survey using name generators asking “who does your household give country food to?” and “who gives your household country food?” Responses to these two questions were aggregated and coded as ties between households. 75% of households in the community responded to the questionnaire; in the data used here, non-responding households are omitted from the network, giving a network of 110 nodes.

As a measure of the resources available for sharing at the household level, we converted household reports of their harvests of four key food species (seal, beluga, geese, and caribou) in the last 12 months to an estimate of calories harvested. The four species reported are not a complete inventory of locally-harvested foods, but we treat them as a reasonable proxy for the total household harvest. Because the harvest size distribution uses these four species, we only keep the edges in the empirical network such that it only retains out-edges from households that reported harvesting from these species. This filtering by itself does not change aggregate network measures in a significant manner, but we include the unfiltered data in our code repository for completeness.

The household foraging returns give us a vector H of length 110 which we pass to the model as the fixed harvest values (resource endowment) of each node. The foraging returns data is zero-inflated, highly skewed and it encompasses a full year of foraging (see Figure S1). For this reason, they are divided by a standard daily caloric intake constant $D = 2500$ and then put through a $\log(1 + x)$ transform¹ to compress the scale, yielding scaled harvest values on the logarithmic scale for each node’s endowment, which are then used in the algorithm that generates communication networks.

To ensure that our agent-based model realistically reproduces the 2013–2014 Kangiqsujuaq sharing network, we calibrate it against carefully chosen network-level summaries. Each summary targets a specific behavioral or structural feature that maps directly onto one or more model mechanisms (see Inference Algorithm).

One of the summary statistics we target, PageRank centrality, requires special explanation. The sharing network data are unweighted: households were asked who their sharing partners are, but not how much or how often they give. Several distinct processes can generate sharing ties of equal weight in this representation, including small gifts that help maintain reciprocal relations, diffuse sharing within kin groups, occasional broadcast giving, and more selective transfers that plausibly function as indirect signals. In this context, we expect raw in-degree centrality to be strongly

¹Base e .

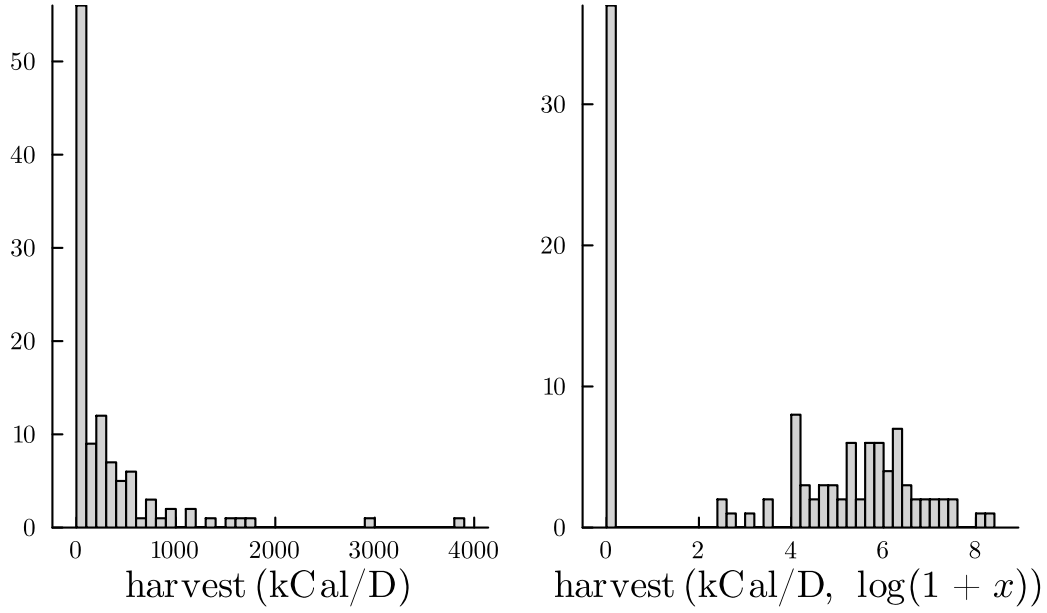


Figure 1: Distribution of year-long household harvest sizes in Kangiqsujuaq for 2013 - 2014. **Left.** Harvest in “person-days” of sustenance, obtained by dividing raw harvest in kilocalories by the daily caloric intake standard $D = 2500$. **Right.** Log-transformed person-day harvest distribution.

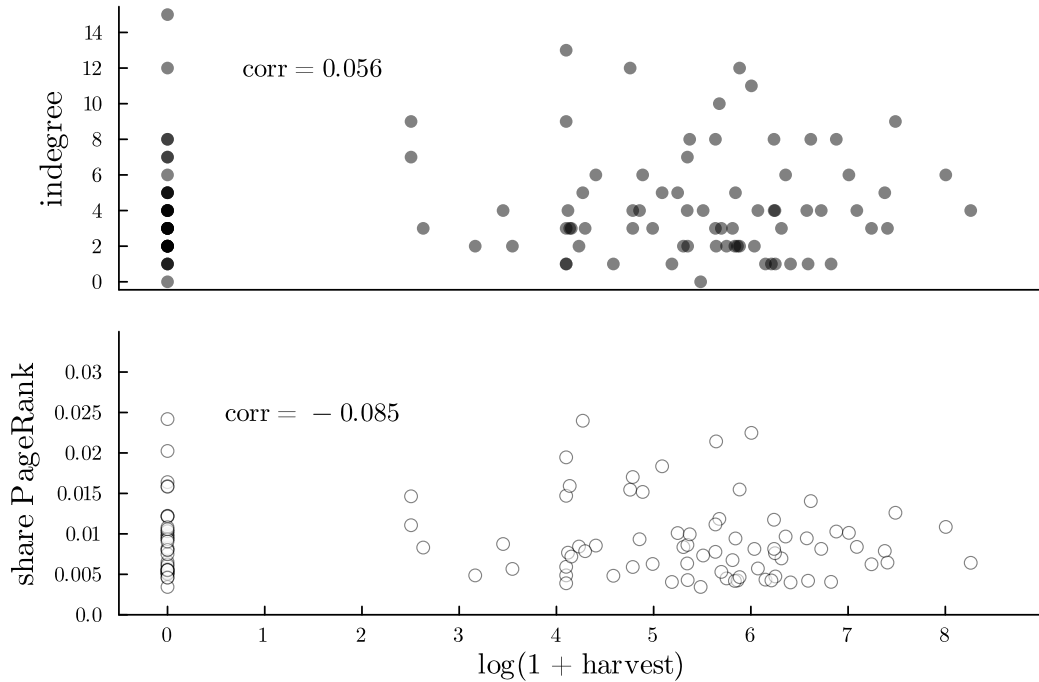


Figure 2: Scatterplots of endowment against in-degree centrality (top) and PageRank centrality (bottom) in the Kangiqsujuaq sharing network, 2013-2014. Spearman correlation coefficients for both joint distributions are printed on their respective scatterplots.

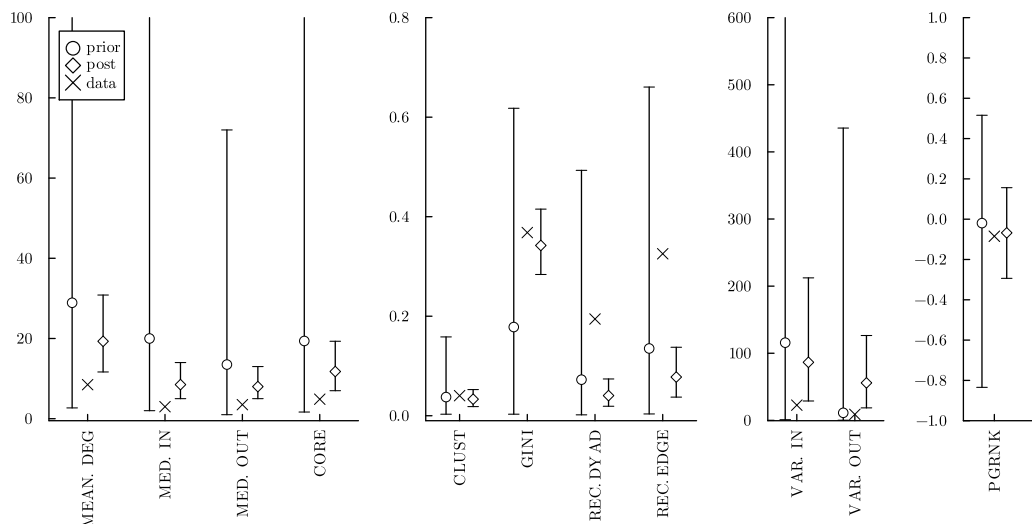


Figure 3: Kernel density plots of the final ABC-PMC marginal pooled posterior for each of the parameters in the strategic network formation simulation model using variance of outdegree instead of MAD of outdegree.

confounded by processes such as direct reciprocity and kin selection, as well as by background “broadcast” sharing.

For this reason, we focus our model fitting on the correlation between PageRank centrality and endowment, rather than on the in-degree–endowment correlation. PageRank simulates a random walk on the sharing graph, so that each incoming tie is weighted by the centrality of the giver. Ties from highly non-selective, high-outdegree givers contribute less marginal PageRank per edge than ties from more selective givers whose out-neighbors are few. Intuitively, PageRank downweights edges from households that “broadcast” to many alters and upweights edges from more selective donors, making it a closer proxy for the selective attention we are interested in when true edge weights are unobserved.

Empirically, endowment is positively correlated with raw sharing in-degree but negatively correlated with PageRank (Figure S2). This joint pattern is consistent with a scenario in which high-harvest households are the focus of much background sharing (inflating their in-degree), while more selective ties that can carry stronger reputational weight concentrate on mid- to lower-harvest households. Because our mechanism is specifically about how reputational incentives shape selective transfers, we therefore use the PageRank–endowment correlation as a key aggregate measure when fitting the simulation model.

4 Alternative PMC fit using outdegree variance

In the main text, we show results for a model fit that uses the median absolute deviation (MAD) of outdegree as a robust measure of outdegree variability, given that the high skewness of the outdegree distribution is driven by a single super-sharer. Supplementary figures 3–6 reproduce the plots in the main text using a model fit using outdegree variance instead of MAD, for comparison. We find that while the inferred posterior distribution of parameters broadly resembles the posterior using MAD, posterior predictions in the outdegree variance model struggle to match the degree distribution of the empirical aggregated dataset compared to the MAD fit.

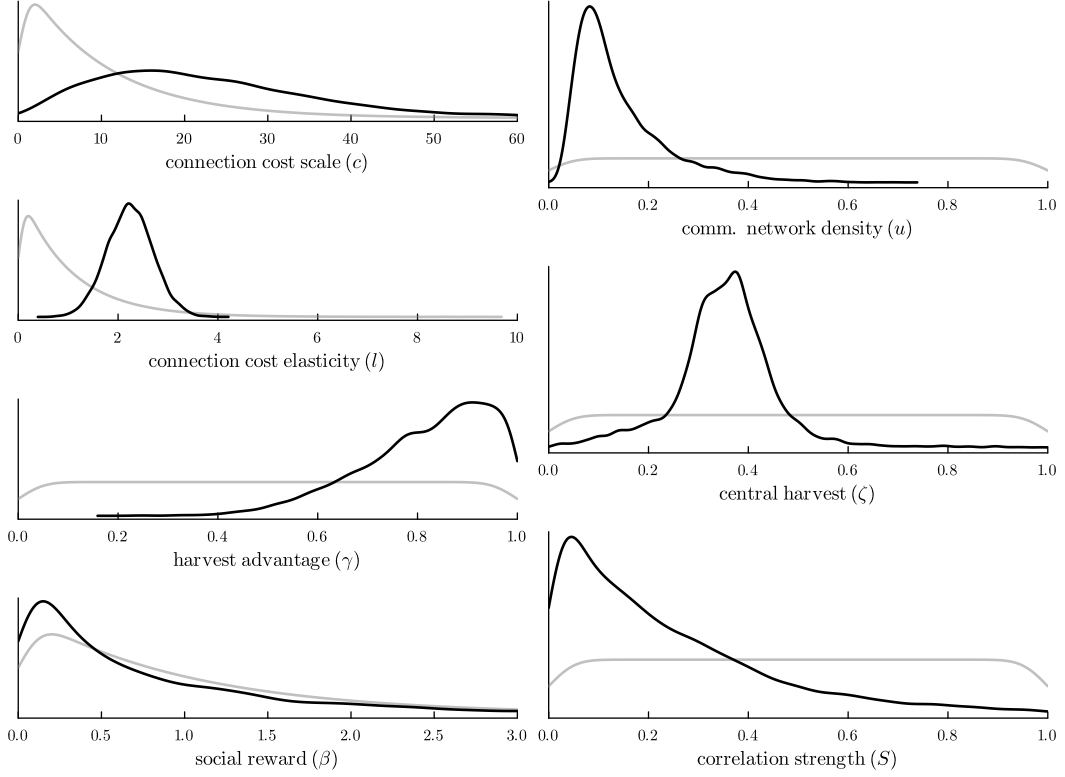


Figure 4: Prior (circle) and posterior (rhombus) predictive checks of the network signaling ABM compared with the empirical sharing network values (cross), showing how the outcome space for several network measures of interest contracts after fitting and how well the model predicts the empirical aggregate network measures, using variance of outdegree instead of MAD of outdegree. The measures are, from left to right: mean degree, median indegree, median outdegree, average core number, average clustering coefficient, Gini coefficient of indegree distribution, dyad reciprocity, edge reciprocity, variance of indegree, variance of outdegree and Spearman correlation between PageRank and harvest.

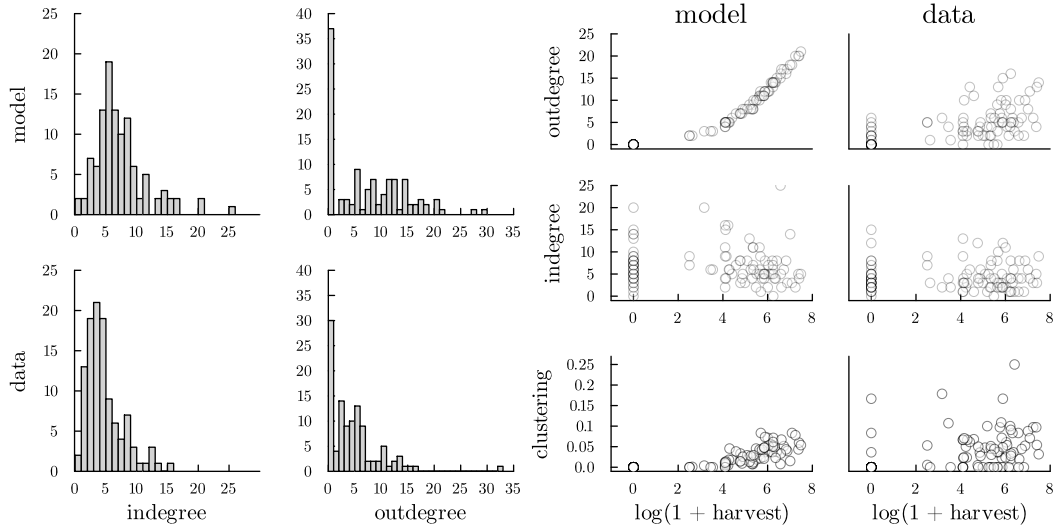


Figure 5: **Left.** Histograms of in- and out-degree distributions in the empirical network versus a posterior simulated network. **Right.** Joint distributions of harvest versus in-degree, out-degree and local clustering coefficient in the empirical network and the posterior simulated network using variance of outdegree instead of MAD of outdegree. Posterior simulations are drawn from the modal parameter combination in the joint pooled ABC-PMC posterior.

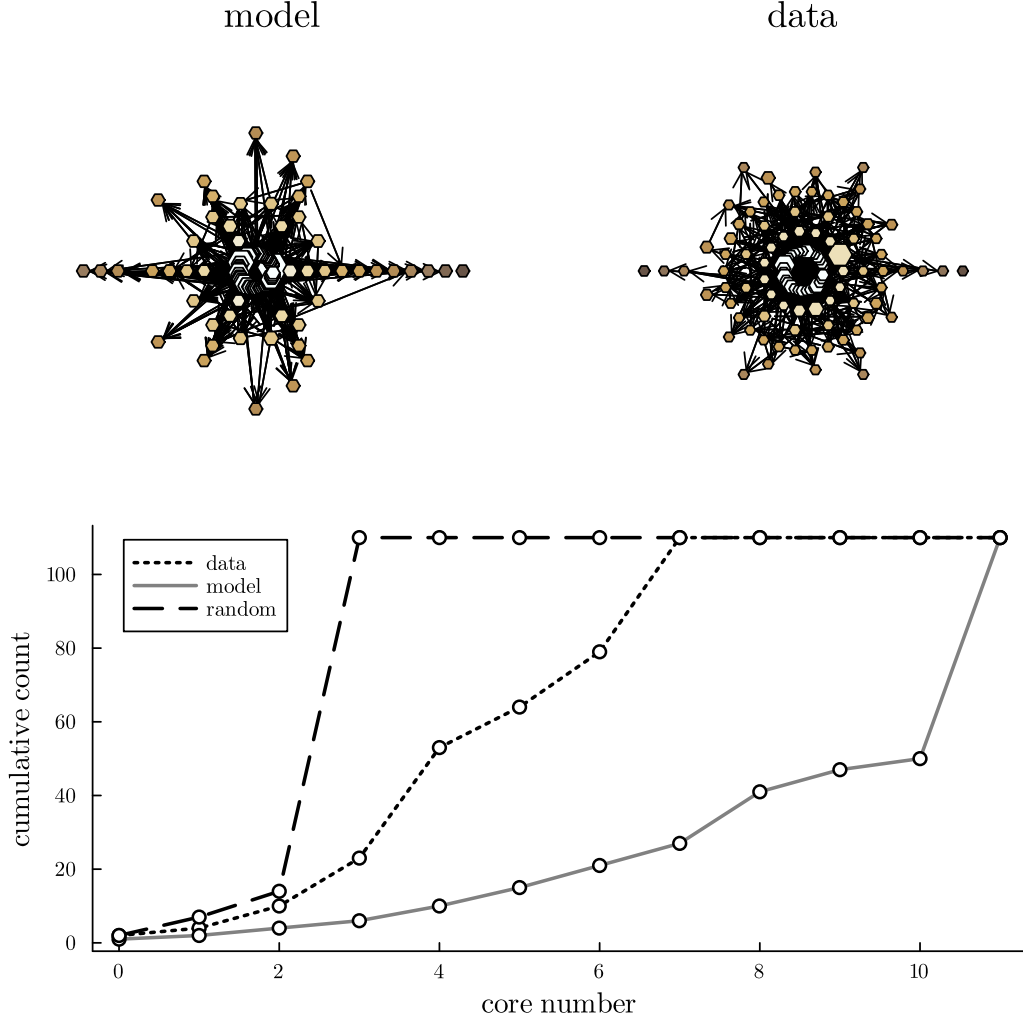


Figure 6: **Top.** Graph visualizations of the coreness structure of the empirical sharing network and a network drawn from the modal parameter combination in the joint pooled posterior using variance of outdegree instead of MAD of outdegree. Nodes are arranged by their k -shell, with higher k -shells being positioned more centrally in the graph. **Bottom.** Cumulative k -core counts across three networks: the empirical Kangiqsujuaq sharing network (dotted), a network sampled from the modal parameter combination in the joint pooled posterior (solid) and an Erdős-Rényi random network with the same density as the empirical network (dashed).

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